Xiang Zheng *"Shee-ahng Jeng"*

xiangzhe@andrew.cmu.edu

x-zhe.github.io

RESEARCH INTERESTS

My research examines how scientific discovery, knowledge production, and research careers are shaped by technological change, institutional policies, and career structures using quantitative methods. I use large-scale publication, funding, peer review, and career data to study innovation, evaluation, inequality, and organizational decision-making in science.

Areas of Specialization: Science of Science; Data Science; Computational Social Science; Information Policy

ACADEMIC APPOINTMENTS

Postdoctoral Fellow Oct 2026–present
Dietrich College of Humanities and Social Sciences, Carnegie Mellon University
Mentor: Dr. Cassidy R. Sugimoto

EDUCATION

Ph.D. in Information Sept 2021 - Sept 2026
Information School, University of Wisconsin-Madison, United States
Core member of the *Metascience Research Lab*

M.S. in Computer Sciences Dec 2023 - Dec 2024
Computer Sciences, University of Wisconsin-Madison, United States

M.A. in Education Aug 2018 - July 2021
School of Education, Tsinghua University, China

Bachelor of Management; Bachelor of Arts (dual degree) Aug 2014 - July 2018
Tsinghua University, China

Direct Enrollment Education Program Oct 2016 - March 2017
Tohoku University, Japan

PUBLICATIONS

Peer-reviewed Journal Articles

$\star$ indicates equal contribution

[J1] Salsabil Arabi, **Xiang Zheng**, B. Ian Hutchins, and Chaoqun Ni. "Equity in Science Journalism: Investigating Gender Disparities in News Media Coverage of Scientific Research". *Science Communication*, 2025. DOI: 10.1177/10755470251360187.
- Featured in ***Science***.

[J2] Kai Li, **Xiang Zheng**, and Chaoqun Ni. "Gender Disparities in the STEM Research Enterprise in China". *Humanities and Social Sciences Communications* 12: 1, 2025. P. 800. DOI: 10.1057/s41599-025-05028-y.

[J3] ⋆Giorgio Tripodi, ⋆**Xiang Zheng**, ⋆Yifan Qian, Dakota Murray, Benjamin F. Jones, Chaoqun Ni, and Dashun Wang. "Tenure and Research Trajectories". *Proceedings of the National Academy of Sciences* 122: 30, 2025. e2500322122. DOI: 10.1073/pnas.2500322122.
• Covered by ***Nature*, *Forbes*, *Inside Higher Ed***, etc.

[J4] **Xiang Zheng** and Chaoqun Ni. "Dissecting the Global Peer Review Ecosystem: Skewness, Homophily, and Diversity Dynamics". *Quantitative Science Studies* 6, 2025. Pp. 463–482. DOI: 10.1162/qss_a_00358.

[J5] **Xiang Zheng**, Qiyao Yang, Jai Potnuri, Chaoqun Ni, and B Ian Hutchins. "Comparing the Outputs of Intramural and Extramural Grants Funded by National Institutes of Health". *eLife* 14, 2025. RP108929. DOI: 10.7554/elife.108929.1.

[J6] ⋆Kai Li, ⋆**Xiang Zheng**, and Chaoqun Ni. "The Anglicization of Science in China". *Research Evaluation* 33, 2024. rvae036. DOI: 10.1093/reseval/rvae036.

[J7] **Xiang Zheng** and Chaoqun Ni. "The Significant yet Short-Term Influence of Research Covidization on Journal Citation Metrics". *Journal of the Association for Information Science and Technology* 75: 9, 2024. Pp. 1002–1017. DOI: 10.1002/asi.24939.

[J8] **Xiang Zheng**, Erjia Yan, and Chaoqun Ni. "Faculty Mobility and Research Dynamics at Historically Black Colleges and Universities". *Quantitative Science Studies*, 2024. Pp. 1–42. DOI: 10.1162/qss_a_00334.

[J9] **Zheng, Xiang**, Wenhui Zhou, Chaoqun Ni, and Chuanyi Wang. "The Influencing Mechanism of Research Training on Chinese STEM Ph.D. Students' Career Interests". *Asia Pacific Education Review* 25: 4, 2024. Pp. 733–749. DOI: 10.1007/s12564-022-09775-4.

[J10] **Xiang Zheng**, Jiajing Chen, Alison Tollas, and Chaoqun Ni. "The Effectiveness of Peer Review in Identifying Issues Leading to Retractions". *Journal of Informetrics* 17: 3, 2023. P. 101423. DOI: 10.1016/j.joi.2023.101423.
• Featured by **Retraction Watch**.

[J11] **Xiang Zheng**, Jiajing Chen, Erjia Yan, and Chaoqun Ni. "Gender and Country Biases in Wikipedia Citations to Scholarly Publications". *Journal of the Association for Information Science and Technology* 74: 2, 2023. Pp. 219–233. DOI: 10.1002/asi.24723.
• Highlighted as **2022 Top Downloaded Paper in *JASIST***.

[J12] **Xiang Zheng**, Shreyas Vastrad, Jibo He, and Chaoqun Ni. "Contextualizing Gender Disparities in Online Teaching Evaluations for Professors". *PLOS ONE* 18: 3, 2023. e0282704. DOI: 10.1371/journal.pone.0282704.

[J13] Jibo He, **Xiang Zheng**, Mingchun Liu, Yu Du, Guanglin Liu, Jingmeng Cui, and Yanjie Su. "Reciprocity in College Teaching: A Big Data Study Based on Online Student Evaluation of 919,750 Professors". *Assessment & Evaluation in Higher Education* 47: 8, 2022. Pp. 1401–1415. DOI: 10.1080/02602938.2022.2067980.

[J14] Xi Hong, Fei Guo, **Xiang Zheng**, and Hamish Coates. "The Meaning of Doctoral Education Quality and Its Evolution". *Modern Education Management*: 04, 2022. Pp. 119–128. DOI: 10.16697/j.1674-5485.2022.04.014.

[J15] **Xiang Zheng**, Haimiao Yuan, and Chaoqun Ni. "Meta-Research: How Parenthood Contributes to Gender Gaps in Academia". *eLife* 11, 2022. e78909. DOI: 10.7554/eLife.78909.
• Covered by ***Chemistry World*, *Women In Academia Report***, etc.

[J16] Jinyuan Ma, Fan Jiang, Liujian Gu, **Xiang Zheng**, Xiao Lin, and Chuanyi Wang. "Patterns of the Network of Cross-Border University Research Collaboration in the Guangdong-Hong Kong-Macau Greater Bay Area". *Sustainability* 12: 17, 2020. P. 6846. DOI: 10.3390/su12176846.

[J17] Chuanyi Wang, **Xiang Zheng**, Yu Zhang, Guannan Yao, and Shining Kang. "Implementation and Diffusion of Online Higher Education Policies —A Case Study on Tsinghua University". *Tsinghua Journal of Education* 41: 04, 2020. Pp. 47–59. DOI: 10.14138/j.1001-4519.2020.04.004713.

[J18] Chuanyi Wang, **Xiang Zheng**, and Jiale Yang. "A Comparative Study on the Development of Postgraduate Education in the Discipline of Education from the Year 2000: Based on the Diachronic Analysis of 8 Countries". *China Higher Education Research*: 07, 2019. Pp. 85–91. DOI: 10.16298/j.cnki.1004-3667.2019.07.13.

[J19] Chuanyi Wang, Jiale Yang, and **Xiang Zheng**. "The Development of Education Discipline in China from the Perspective of SWOT". *Modern Education Management*: 03, 2019. Pp. 1–6. DOI: 10.16697/j.cnki.xdjygl.2019.03.001.

Book Chapters and Translation

[C1] **Xiang Zheng**. "Educational Modernization in Japan". In: Huiqin Liu. *International Comparison of Educational Modernization*. Beijing: People's Education Press.
- In press.

[C2] Blaise Cronin and Cassidy R Sugimoto, eds. *Beyond Bibliometrics: Harnessing Multidimensional Indicators of Scholarly Impact (Chinese ver.)* Trans. by **Xiang Zheng**, Chuanyi Wang, and Han Chu. Beijing: China Science and Technology Press, 2023.

[C3] Chuanyi Wang and **Xiang Zheng**. "Enacting Collective and Inquiry-Based Leadership". In: *A Turning Point for Chinese Higher Education*. Routledge, 2022. ISBN: 978-1-00-325775-2. DOI: 10.4324/9781003257752.

Working Papers and Manuscripts

[M1] Peiran Li, Fangzhou Lin, Shuo Xing, **Xiang Zheng**, Xi Hong, Siyuan Yang, Jiashuo Sun, Zhengzhong Tu, and Chaoqun Ni. *BibAgent: An Agentic Framework for Traceable Miscitation Detection in Scientific Literature*. Jan. 30, 2026. DOI: 10.48550/arXiv.2601.16993.

[M2] Erjia Yan, Chaoqun Ni, and **Xiang Zheng**. *Global Training and the Collaborative Structure of Elite U.S. Science*. May 18, 2026. DOI: 10.48550/arXiv.2605.18715.

[M3] Xiang Zheng, Xi Hong, Jialin Liu, and Chaoqun Ni. *Scientific Exploration, Collaboration and Labor Division in the Large Language Model Era*. July 23, 2026. DOI: 10.48550/arXiv.2607.20923.
- Poster accepted in NeurIPS 2026 Workshop on AI-Native Academia.

[M4] Xi Hong, **Xiang Zheng**, Haimiao Yuan, and Chaoqun Ni. *Parenthood Penalties in Academia: Childcare Responsibilities, Gender Role Beliefs and Institutional Support*. Apr. 11, 2025. DOI: 10.48550/arXiv.2504.13923.
- Under revision in *Humanities and Social Sciences Communications*.

[M5] **Xiang Zheng**, Anli Peng, Xi Hong, and Chaoqun Ni. *Interdisciplinary PhDs Face Barriers to Top University Placement within Their Disciplines*. Mar. 27, 2025. DOI: 10.48550/arXiv.2503.21912.
- Under revision in *Nature Communications*; **Best Speaker Award at ICSSI 2026**.

[M6] Kyrie Zhixuan Zhou, Justin Eric Chen, **Xiang Zheng**, Yaoyao Qian, Yunpeng Xiao, and Kai Shu. *"Everyone Else Does It": The Rise of Preprinting Culture in Computing Disciplines*. Nov. 6, 2025. DOI: 10.48550/arXiv.2511.04081.
- Accepted in 2026 ASIS&T.

CONFERENCE ACTIVITIES

Refereed Oral Presentations

[O1] "Interdisciplinary Ph.D. s Face Barriers to Top University Placement". ICSSI 2026 (International Conference on the Science of Science & Innovation) (**Best Speaker Award)**.

[O2] "Equity in Science Journalism: Investigating Gender Disparities in News Media Coverage of Scientific Research". ATLC 2025 (Atlanta Conference on Science and Innovation Policy); IC2S2 2025 (International Conference on Computational Social Science).

[O3] "Interdisciplinary PhDs Face Barriers to Top University Placement within Their Disciplines". ATLC 2025; IC2S2 2025.

[O4] "Parenthood Penalties in Academia: Childcare Responsibilities, Gender Role Beliefs and Institutional Support". IC2S2 2025.

[O5] "Contextualizing Gender Disparities in Online Teaching Evaluations for Professors". IC2S2 2024.

[O6] "Tenure and Research Trajectories". ICSSI 2024; IC2S2 2024; NBER Advancing the Science of Science Funding Workshop 2025.

[O7] "Skewness and Homophily in the Global Peer Review Network". ISSI 2023 (International Society for Scientometrics and Informetrics).

[O8] "The anglicization of Science in China". ISSI 2023.

[O9] "The Effectiveness of Peer Review in Identifying Issues Leading to Retractions". STI 2023 (International Conference on Science, Technology and Innovation Indicators).

[O10] "The Significant yet Short-Term Influence of Research Covidization on Journal Citation Metrics". ICSSI 2023.

[O11] "How Parenthood Contributes to Gender Gaps in Academia". ICSSI 2022.

[O12] "Understanding Gender Disparity in Publication by Scientists' Work-family Conflict, Social Support, and Gender Role Belief". CIES 2022 (Comparative & International Education Society annual conference).

Refereed Posters

[P1] "How Do Academic Gatekeepers Evaluate Figures? An Empirical Study of Peer Review Reports Using Large Language Models". ICSSI 2026.

[P2] "Scientific Exploration, Collaboration and Labor Division in the Large Language Model Era". NeurIPS 2026 Workshop on AI-Native Academia Accepted poster.

[P3] "Evaluating the Language Style of AI-Generated Peer Review". iConference 2025 **(Best Poster Finalist)**.

[P4] "Analysis of the Comparative Strengths of Intramural and Extramural Grant Funding Mechanisms". ICSSI 2024.

[P5] "The Tenure Debate: How US Faculty Change Their Research Practices Post-Tenure". iConference 2024.

[P6] "Gender Disparities in the STEM Research Enterprise of China". ICSSI 2023.

Program Committee and Session Service

Program Committee member
Wiki Workshop 2025, ATLC 2025

Chair
AI in Education Session, IC2S2 2024

INVITED TALKS

Beyond the Tenure Debate: What do faculty career and research trajectories show? (2026). School of Information, University of California, Berkeley, United States.

Tenure and Research Trajectories (2025). Institute for Humane Studies (IHS) December Graduate Conference, George Mason University, United States.

Data Visualization using Tableau and Python (2025). Information School, University of Wisconsin-Madison, United States.

Tenure and Research Trajectories in the US (2024). Department of Statistics and Data Science, Southern University of Science and Technology, China.

Tenure and Research Trajectories in the US (2024). Academic Analytics Research Center Webinar Series (virtual).

Exploring the impact of tenure on research performance, innovation, and collaboration (2024). The 5th annual Research Bazaar, University of Wisconsin-Madison, United States.

Frontiers in Teaching Evaluation and Research Evaluation Based on Scientific Metrics (2023). Institute of Education, Tsinghua University, China.

AWARDS AND HONORS

Best Speaker Award June 2026
ICSSI 2026: the International Conference on the Science of Science & Innovation

Nominee for the 2027 Schmidt Science Fellows May 2026
University of Wisconsin-Madison

Graduate Student Teaching Award April 2026
Information School, University of Wisconsin-Madison

Best Poster Finalist March 2025
iConference 2025, Indiana University Bloomington

Humane Studies Fellowship ($5000) Dec 2024
Institute for Humane Studies at George Mason University

Travel Funding Award ($600) June 2023, July 2024
ICSSI 2023, ICSSI 2024: the International Conference on the Science of Science & Innovation

Student Research Grants Competition Award ($600) May 2023
University of Wisconsin-Madison

1st Prize in Comprehensive Excellent Scholarship Dec 2019
Tsinghua University

TEACHING EXPERIENCE

Lecturer (Primary Instructor)

LIS 407 Data Storytelling with Visualization Summer 2024 - Summer 2026
Information School, University of Wisconsin-Madison.
Seven consecutive semesters; in-person and online instruction; Python, Tableau, Gephi.

Teaching Assistant

LIS 351 Introduction to Digital Information Spring 2022
Information School, University of Wisconsin-Madison. Instructor: Corey Jackson
LIS 706 Data Mining Planning and Management Fall 2021
Information School, University of Wisconsin-Madison. Instructor: Ian Hutchins

Research Mentoring

Baolu Yu, Information PhD. Topic: scientific mobility.
Salsabil Arabi, Information PhD. Topic: Bias in news media.
Shreyas Vastrad, Information MS. Topic: NLP analysis.
Xiaopeng Yuan, Information MS. Topic: Race prediction by machine learning.
Jiajing Chen, CS MS at New York University. Topic: peer review bias.
Qiyao Yang, CS undergraduate. Topic: research funding efficiency.
Peiran Li, CS undergraduate. Topic: Miscitation identification using LLM.

SERVICE TO PROFESSION

Journal Manuscript Review

Interdisciplinary: *Science* (2026); *Nature Human Behaviour* (2026); *Nature Communications* (2026); *Science Advances* (2026); *Communication Medicine* (2026); *Scientific Data* (2025); *PLOS One* (2024); *Humanities & Social Sciences Communications* (2022).

Information Science: *Scientometrics* (2026); *Journal of Medical Internet Research* (2026); *Information Processing and Management* (2025); *Journal of the Association for Information Science and Technology (JASIST)* (2024); *Frontiers in Research Metrics and Analytics* (2023); *Journal of Information Science* (2022); *Publications* (2022).

Education: *Education Sciences* (2024); *MedEdPublish* (2024); *Trends in Higher Education* (2024); *International Journal of Chinese Education* (2022); *Teaching of Psychology* (2021).

General Social Sciences: *Discover Global Society* (2026); *Family Relations* (2026); *Economies* (2025); *Genealogy* (2025); *Psychology of Language and Communication* (2025); *Social Currents* (2024); *Journal of Accounting Literature* (2024); *Journal of Crisis Intervention and Suicide Prevention* (2023); *Sustainability* (2023).

Conference Manuscript Review

iConference	2027
NeurIPS 2026 Workshop on AI-Native Academia	2026
Wiki Workshop	2025, 2026
International Conference for Computational Social Science (IC2S2)	2025
Atlanta Conference on Science and Innovation Policy (ATLC)	2024
ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW)	2024
International Conference on Science, Technology and Innovation Indicators (STI)	2023
Comparative & International Education Society annual conference (CIES)	2023

SELECTED MEDIA COVERAGE

[1] "Time Well Spent: Productivity with Purpose in Research". *Spectroscopy Online*, 2025.

[2] Max Kozlov. "How Getting Tenure Changes Researchers' Publication Habits — and Citations". *Nature* 643: 8074, 2025. Pp. 1165–1166.

[3] Anirban Mukhopadhyay. "When Women Researchers Publish, Media Attention Doesn't Always Follow". *Science*, 2025.

[4] Michael T. Nietzel. "The Impact Of Faculty Tenure On Research Output Explored In New Study". *Forbes*, 2025.

[5] Kathryn Palmer. "Study: Tenure Doesn't Slow Average Research Output". *Inside Higher Ed*, 2025.

[6] "Gender and Country Biases in Wikipedia Citations to Scholarly Publications". *Wikimedia Research Showcase*, 2023.

[7] "How Parenthood Contributes to Gender Gaps in Academia". *Dementia Researcher*, 2023.

[8] "How Retracted Articles Survived Peer Review". *Revista Pesquisa Fapesp*, 2023.

[9] "Study Examines the "Parent Penalty" for Women in Academia". *Women In Academia Report*, 2022.

[10] Maria Burke. "Parenting Takes a Heavy Toll on the Careers of Female Academics in North America". *Chemistry World*, 2022.